

PRODUCT REQUIREMENTS DOCUMENT (PRD)

AWS Operations Intelligence AI

Enterprise Operations Control Center (EOCC)

Version: 1.0

Executive Summary

Modern enterprises have invested millions of dollars in operational tooling:

- AWS
- Splunk
- Jira
- ServiceNow
- Confluence
- GitHub
- Microsoft Teams
- Internal Runbooks
- Architecture Documentation

However, operational teams still face the same challenge every day:

Critical information exists.

Critical intelligence does not.

When an incident occurs, engineers must manually gather information from multiple disconnected systems before they can begin solving the actual problem.

This results in:

- Long investigation times
- Slow incident resolution
- Repeated incidents
- Tribal knowledge dependency
- Ownership confusion
- Operational inefficiency
- Increased business risk

Current platforms monitor systems.

Current platforms do not operate the enterprise.

AWS Operations Intelligence AI is designed to become the Enterprise Operations Control Center.

The platform continuously monitors, understands, governs, coordinates, investigates, and assists operational activities across the enterprise.

Instead of being another monitoring tool, the platform acts as an intelligent operational control layer that sits above existing enterprise systems.

Vision

Create the operational brain of the enterprise.

A continuously running intelligence platform capable of understanding:

- Applications
- AWS Services
- Teams
- Business Processes
- Ownership
- Dependencies
- Operational Knowledge
- Historical Incidents
- Architecture
- Risk

and transforming that understanding into actionable operational intelligence.

Product Mission

Transform enterprise operations from:

Reactive Operations

to

Intelligent Operations

and eventually

Autonomous Operations

through a continuously learning agentic platform.

Product Positioning

We are NOT building:

- Observability Platform
- Monitoring Platform
- Logging Platform
- Dashboard Platform
- Chatbot

These solutions already exist.

We ARE building:

Enterprise Operations Control Center

A centralized operational intelligence layer that coordinates enterprise operations using AI.

Core Problem Statement

Every enterprise suffers from the same operational challenges:

Problem 1

Information Fragmentation

Knowledge exists across:

- AWS
- Splunk
- Jira
- ServiceNow
- Confluence
- Teams
- GitHub
- Runbooks

No single system understands everything.

Problem 2

Operational Tribal Knowledge

Critical knowledge exists only inside employee heads.

Organizations become dependent on specific individuals.

Problem 3

Ownership Visibility

Teams struggle to identify:

- Service owners
 - Business owners
 - Escalation paths
 - Support teams
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Problem 4

Dependency Blindness

Organizations do not fully understand:

- Application dependencies
- Team dependencies
- Business dependencies

until failures occur.

Problem 5

Repeated Incident Cycles

Organizations repeatedly solve the same incidents.

Institutional learning is lost.

Problem 6

Business Impact Uncertainty

Most operational tools answer:

"What failed?"

Few answer:

"What business capability is impacted?"

Enterprise Operations Control Center

The platform operates through five core pillars.

Pillar 1

Enterprise Monitoring

Continuously observe:

AWS Services

- ECS
- Lambda
- Glue
- Step Functions
- RDS
- DynamoDB
- SQS
- SNS
- CloudWatch

Operational Systems

- Splunk

Process Systems

- Jira
- ServiceNow

Knowledge Systems

- Confluence
- SharePoint
- Runbooks

Communication Systems

- Microsoft Teams

Development Systems

- GitHub

The platform creates a unified operational view.

Pillar 2

Enterprise Intelligence

Build and maintain a living Enterprise Intelligence Graph.

Relationships include:

Business Process ↓ Application ↓ AWS Service ↓ Owner Team ↓ Manager ↓ Documentation ↓ Runbook
↓ Historical Incidents ↓ Dependencies

The graph becomes the operational knowledge foundation.

Pillar 3

Enterprise Investigation

AI agents automatically investigate:

- Failures
- Service degradation
- Operational anomalies
- Dependency issues
- Ownership conflicts

Investigation reports are generated automatically.

Pillar 4

Enterprise Coordination

AI coordinates:

- Incident response
- Team notifications
- Escalations
- Ownership assignment
- Follow-up activities

The platform acts as an operational coordinator.

Pillar 5

Enterprise Control

The platform supports controlled operational execution.

Examples:

- Restart ECS Services
- Trigger Glue Jobs
- Execute Runbooks
- Create Jira Tickets
- Notify Teams
- Trigger Recovery Workflows

Human approval remains in control.

Agentic Architecture

The platform operates through specialized AI agents.

Ownership Agent

Responsible for:

- Ownership discovery

- Escalation path identification
 - Team mapping
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Dependency Agent

Responsible for:

- Service relationships
 - Application relationships
 - Team relationships
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Incident Agent

Responsible for:

- Incident investigation
 - Historical incident analysis
 - Resolution recommendations
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Knowledge Agent

Responsible for:

- Documentation retrieval
 - Runbook understanding
 - Knowledge synthesis
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Governance Agent

Responsible for:

- Ownership gaps
 - Documentation gaps
 - Operational maturity analysis
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Control Agent

Responsible for:

- Controlled operational actions
 - Runbook execution
 - Recovery orchestration
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Top 20 Business-Critical Use Cases

1. Enterprise Ownership Intelligence

Instantly identify service ownership and escalation paths.

Business Value: Reduce ownership confusion.

2. Business Process Monitoring

Monitor business capabilities instead of infrastructure metrics.

Examples:

- Customer Onboarding
- Order Processing
- Payment Processing

Business Value: Align operations with business outcomes.

3. Enterprise Dependency Intelligence

Understand impact propagation across applications and teams.

Business Value: Reduce investigation time.

4. AI Incident Investigation

Automatically investigate operational failures.

Business Value: Reduce manual effort.

5. Enterprise Knowledge Search

Unified search across all enterprise knowledge.

Business Value: Faster access to institutional knowledge.

6. Historical Incident Intelligence

Retrieve previous incident resolutions.

Business Value: Prevent repeated investigations.

7. Autonomous Incident Coordination

Coordinate teams automatically during incidents.

Business Value: Improve response efficiency.

8. AI Executive Briefings

Generate business-focused incident summaries.

Business Value: Leadership visibility.

9. Architecture Intelligence

Provide architecture understanding on demand.

Business Value: Accelerate onboarding and troubleshooting.

10. Change Risk Intelligence

Assess deployment risks before release.

Business Value: Reduce production incidents.

11. Operational Risk Detection

Identify enterprise operational risks.

Business Value: Prevent outages.

12. Ownership Gap Detection

Identify services without owners.

Business Value: Improve accountability.

13. Runbook Coverage Analysis

Identify missing operational procedures.

Business Value: Improve resiliency.

14. Documentation Governance

Detect outdated documentation.

Business Value: Improve knowledge quality.

15. Organizational Dependency Analysis

Understand cross-team dependencies.

Business Value: Improve operational planning.

16. Critical Expert Detection

Identify organizational dependency on individuals.

Business Value: Reduce key-person risk.

17. Service Consumption Intelligence

Track AWS service consumers.

Business Value: Improve governance.

18. Enterprise Operational Search

Single search interface across enterprise systems.

Business Value: Reduce information retrieval effort.

19. Operational Digital Twin

Create a live enterprise operational model.

Business Value: Improve situational awareness.

20. Controlled Operational Execution

Allow approved actions directly from the platform.

Examples:

- Restart Services
- Execute Runbooks
- Trigger Recovery Workflows

Business Value: Reduce recovery time.

Differentiators

Traditional Tools Focus On:

- Metrics
- Logs
- Alerts
- Dashboards

AWS Operations Intelligence AI Focuses On:

- People
 - Teams
 - Ownership
 - Business Processes
 - Knowledge
 - Dependencies
 - Governance
 - Operational Control
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Success Metrics

Operational Metrics

- 80% reduction in investigation effort
- 50% reduction in incident resolution time
- 60% reduction in knowledge search time

Business Metrics

- Reduced downtime
- Improved operational maturity
- Reduced dependency on tribal knowledge
- Improved governance

Organizational Metrics

- Increased ownership coverage
 - Increased documentation quality
 - Increased runbook coverage
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Long-Term Vision

The Enterprise Operations Control Center becomes the central operating system for enterprise operations.

Instead of opening multiple tools to understand what is happening, why it is happening, who is responsible, and what should be done next, employees interact with a single intelligent platform capable of monitoring, understanding, coordinating, governing, and controlling enterprise operations 24 hours a day, seven days a week.

The platform evolves from an operational assistant into a trusted operational partner capable of continuously improving enterprise reliability, efficiency, governance, and business resilience.